The Impact of Suburb-Level Encoding on Melbourne Housing Price Prediction

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Abstract

This paper examines whether suburb-level encoding improves housing price prediction on a Melbourne housing dataset. The workflow covers data cleaning, missing-value handling, categorical encoding, outlier treatment, exploratory analysis, and regression modeling. We evaluate Linear Regression, Ridge Regression, Lasso Regression, Random Forest, Extra Trees, XGBoost, and LightGBM using 5-fold cross-validation with log-transformed house prices as the target.

Among the evaluated models, XGBoost and LightGBM achieved the best predictive performance, both reaching an R^2 of 0.843 on the log-price scale. To further examine the role of locality, we conducted a locality ablation and leakage-control study using suburb encoding. Adding suburb information improved hold-out performance, while a shuffled-suburb control degraded results, indicating that the gain comes from meaningful locality signal rather than spurious encoding effects. SHAP analysis further showed that `Suburb`, `Distance`, `Landsize`, and `Rooms` were the most influential features. These results show that both model choice and fine-grained locality encoding are important for housing price prediction.

Keywords: Housing price prediction; Melbourne housing market; locality encoding; gradient boosting; interpretable machine learning

1 Introduction

Accurate housing price prediction remains an important problem in real estate analytics because residential values are influenced by several interacting factors, including

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structural characteristics, accessibility, neighborhood context, and broader market conditions [2, 3]. Reliable prediction models can support pricing decisions, investment assessment, and market analysis. At the same time, housing data are difficult to model well because relationships among variables are often nonlinear and because price patterns can differ substantially across localities [4, 5].

In this study, we use a Melbourne residential property dataset containing transaction records and property attributes such as the number of rooms, bathrooms, car spaces, land size, distance from the central business district, property type, and regional location. In addition to these structural variables, the dataset also includes suburb and region information, which makes it suitable for studying how much detailed locality information contributes to prediction performance.

One of the main questions in this work is whether suburb-level information adds predictive value beyond structural and region-level features. While many studies include general location variables, fewer works explicitly test the incremental effect of fine-grained locality encoding under a consistent experimental setting. This is the main focus of our paper. We also compare linear, bagging, and boosting models to examine which type of model is better suited to this dataset. Linear models offer simplicity and interpretability, whereas tree-based models are often better at capturing nonlinear interactions among features [7, 6].

To address these questions, we build an applied machine learning pipeline for Melbourne housing price prediction. The pipeline covers data cleaning, missing-value handling, categorical encoding, outlier treatment, exploratory data analysis, and model comparison. Seven regression models are evaluated: Linear Regression, Ridge Regression, Lasso Regression, Random Forest, Extra Trees, XGBoost, and LightGBM. Model performance is first compared using 5-fold cross-validation. We then conduct a locality ablation and leakage-control study to examine the contribution of suburb-level encoding and to test whether its observed gain persists when the locality labels are disrupted.

The main contributions of this study are as follows:

- We develop a complete applied machine learning pipeline for Melbourne housing price prediction, covering data preparation, exploratory analysis, feature transformation, model comparison, and evaluation.
- We provide an empirical comparison across linear, bagging, and boosting regressors under a common experimental setting, including Linear Regression, Ridge Regression, Lasso Regression, Random Forest, Extra Trees, XGBoost, and LightGBM.
- We conduct a focused locality ablation study and show that suburb-level encoding yields measurable improvements in predictive performance.
- We quantify leakage prevention through a shuffled-suburb control and use SHAP analysis to provide model-level interpretability beyond standard feature importance.

The remainder of this paper is organized as follows. Section 2 reviews prior work on housing price prediction and location-aware real estate modeling. Section 3 presents the dataset, preprocessing procedures, feature construction, and experimen-

tal setup. Section 4 reports the empirical results and discusses their implications. Section 5 concludes the paper and outlines possible future work.

2 Related Work

2.1 Housing Price Prediction and Hedonic Modeling

Housing price analysis has long been studied through hedonic pricing models, where property value is expressed as a function of structural, locational, and environmental characteristics [8, 9]. In this framework, variables such as floor area, number of rooms, accessibility, and neighborhood characteristics are commonly used to explain market value. These models provide a strong conceptual basis for understanding the determinants of housing prices, but their predictive performance can be limited when the underlying relationships are nonlinear or involve complex interactions.

2.2 Machine Learning Approaches for Real Estate Valuation

More recent studies have applied machine learning techniques to housing price prediction, including linear regression, regularized regression, decision trees, random forests, gradient boosting, and neural networks [4, 5, 18]. Compared with classical econometric approaches, machine learning models are often better able to capture nonlinear relationships and interaction effects, especially when the data contain a mixture of numerical and categorical variables. Among these techniques, ensemble tree models have frequently shown strong performance on tabular real estate datasets because of their flexibility and robustness [10, 11].

Linear and regularized linear models nevertheless remain important baselines because they are computationally efficient, relatively interpretable, and still widely used in applied regression tasks [12, 13]. For this reason, comparing linear baselines with a tree-based ensemble model remains useful, especially when the goal is to understand which assumptions fit a particular local housing market more closely. More recent review and interpretability-focused studies have also emphasized the growing role of machine learning in real estate research and the importance of transparent predictive modeling [21, 22, 23].

2.3 Role of Geographic and Locality Information

Location is consistently recognized as one of the most important determinants of housing value [8, 14]. Earlier studies incorporated geographic context through features such as distance to the city center, administrative region, latitude and longitude,

or neighborhood indicators. However, the granularity of location encoding differs considerably across datasets and modeling pipelines. Coarse regional features can capture broad spatial patterns, whereas finer locality variables may better reflect neighborhood-level differences in accessibility, amenities, prestige, and market segmentation [15, 11].

In practical machine learning workflows, high-cardinality location variables such as suburb or neighborhood often require dedicated encoding strategies. Target encoding is one commonly used approach because it preserves category-specific signal while avoiding the sparsity created by high-dimensional one-hot representations [16, 17]. Motivated by this idea, the present study explicitly evaluates whether suburb-level encoding improves housing price prediction beyond broader structural and regional attributes in the Melbourne housing market.

Taken together, the prior literature suggests that both model choice and geographic representation matter, but the incremental value of suburb-level encoding remains underexplored under a consistent experimental setting. The next section therefore describes the dataset, preprocessing steps, feature construction, and evaluation setup used to examine this question.

3 Methodology

This section describes the dataset, preprocessing steps, feature design, and experimental setup used in this study.

3.1 Dataset Description

This study uses the Melbourne Housing Snapshot dataset from Kaggle [19], with additional project-level reference to a related public GitHub implementation [20]. The dataset contains residential property transactions and associated property attributes. It includes structural, transactional, and geographic variables relevant to real estate valuation, such as the number of rooms, transaction price, property type, sale date, distance from the central business district, region name, number of bedrooms, number of bathrooms, number of car spaces, and land size. The objective is to predict residential property prices from these explanatory variables.

Because housing prices typically exhibit a right-skewed distribution, the prediction target is defined as the natural logarithm of the transaction price. This transformation reduces skewness and stabilizes variance, which makes the regression task more stable.

3.2 Data Cleaning and Preprocessing

The raw dataset initially contains 13,580 observations. Data preprocessing was carried out in several stages.

First, attributes that were not retained for predictive modeling were removed. These include `Address`, `Propertycount`, `Method`, `SellerG`, `Postcode`, and `CouncilArea`. In addition, `BuildingArea` and `YearBuilt` were excluded because of a high proportion of missing values. The remaining rows with a small amount of missing data were then removed.

Second, invalid observations were filtered out. Records with nonpositive values in key numerical variables such as `Rooms`, `Price`, and `Landsize` were removed. Duplicate entries were also identified and discarded.

Third, several count-based variables were converted to integer type, and the `Date` field was parsed into datetime format. Additional temporal attributes, namely year and month, were extracted from the transaction date for subsequent analysis.

Fourth, categorical location variables were processed to reduce sparsity. Rare suburbs with no more than 10 transactions were grouped into a common `Other` category. In addition, very small region categories were removed because of insufficient sample size. For categorical encoding, `Type` and `Regionname` were transformed using one-hot encoding, while `Suburb` was reserved for target encoding during model training.

Finally, outlier treatment was applied after defining the transformed target

$$y = \log(\texttt{Price}).$$

A z-score-based filtering procedure was applied to `log_price`, `Rooms`, `Bedroom2`, `Bathroom`, and `Car`. An interquartile range (IQR) based procedure was applied to `Distance` and `Landsize`. After preprocessing and outlier treatment, the cleaned dataset contained 10,480 observations.

3.3 Exploratory Data Analysis

Exploratory data analysis was conducted to identify preliminary relationships between explanatory variables and housing prices. The analysis showed that structural variables such as the number of rooms, bedrooms, bathrooms, and land size had clearer relationships with log-transformed prices than with raw prices. In addition, distance from the central business district showed a negative relationship with property value, while comparisons across regions and suburbs suggested that geographic location plays an important role in price variation.

Figure 1 compares the target distribution before and after logarithmic transformation and shows that the log transformation substantially reduces skewness. This supports the use of `log_price` as a more stable target for regression. The correlation structure shown in Figure 2 further highlights the positive association of rooms,

bathrooms, and land size with price, and the negative association of distance with price.

These observations motivated two modeling decisions. First, log-transformed price was adopted as the prediction target. Second, the contribution of suburb-level information was explicitly tested through dedicated ablation and robustness experiments.

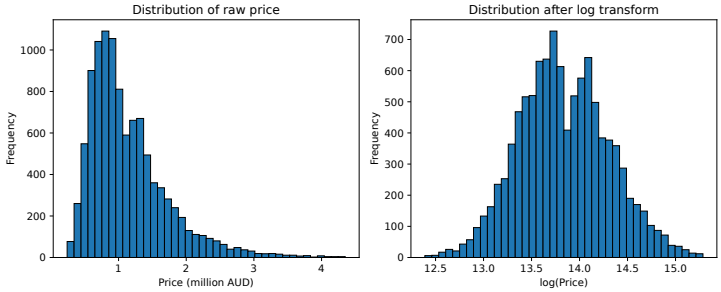


Fig. 1 Distribution of the target before and after logarithmic transformation. The log transformation reduces skewness and makes the regression target more stable.

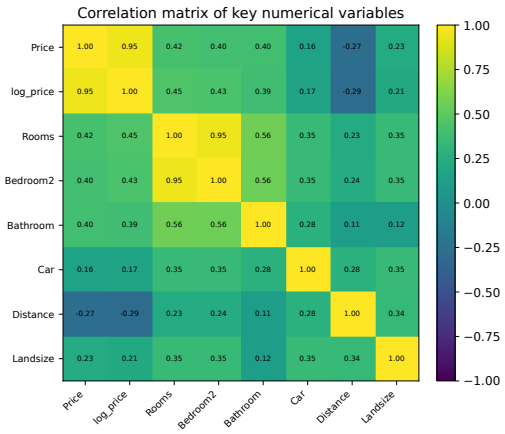


Fig. 2 Correlation matrix of key numerical variables used during exploratory analysis. The figure highlights the strong positive associations of rooms, bathrooms, and land size with housing prices, and the negative association of distance with price.

3.4 Feature Construction

The main prediction target in this study is `log_price`. The baseline feature groups used in the experiments are summarized in Table 1.

Table 1 Feature groups used in the experiments.

Feature group	Variables	Baseline	With suburb
Structural	Rooms, Bathroom, Car, Landsize, Distance	Yes	Yes
Property type	Type_h, Type_t, Type_u	Yes	Yes
Region	Region_EM, Region_NM, Region_SEM, Region_SM, Region_WM	Yes	Yes
Locality	Suburb (target encoded)	No	Yes

To evaluate the predictive value of fine-grained locality information, a second feature setting was constructed by additionally incorporating `Suburb` through target encoding. This resulted in two experimental conditions:

- models trained without suburb-level information, and
- models trained with suburb-level information.

This design allows us to assess the incremental contribution of suburb-level encoding beyond broader structural and regional attributes.

3.5 Regression Models

Seven regression models were evaluated, as summarized in Table 2.

Table 2 Regression models evaluated in the study.

Linear Regression	Ridge Regression	Lasso Regression	Random Forest
Extra Trees	XGBoost	LightGBM	–

For the linear models, target encoding was applied to `Suburb`, followed by feature standardization using `StandardScaler`. For the tree-based and boosting models, target encoding was applied without standardization, since these models are not sensitive to feature scale in the same manner as linear models. To reduce the risk of leakage and overfitting, target encoding was fitted only on the training portion of each split, with `smoothing`=10 and `min_samples_leaf`=20. In the cross-validation setting, the encoder was fitted separately within each fold. In the final train/test experiments, it was fitted on the training split and then applied to the test split.

For the Random Forest locality ablation study, the final configuration used 300 trees with `max_depth`=12, `min_samples_leaf`=8, and `min_samples_split`=20. For the boosting baselines, XGBoost used 300 estimators with `max_depth`=6 and `learning_rate`=0.05, while LightGBM used 300 estimators with `num_leaves`=31 and `learning_rate`=0.05.

3.6 Experimental Setup and Evaluation Metrics

Model development was based on an 80/20 train–test split, while model comparison was conducted using 5-fold cross-validation on the training set. The evaluated performance metrics were Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). For the extended baseline comparison, cross-validation results are reported as mean ± standard deviation across folds.

All model comparison and robustness results in the final revised version are reported on the log-price scale in order to provide a consistent comparison across models trained on the transformed target. In addition to the main model comparison, we performed a shuffled-suburb control and inspected the learning curve of the final pipeline to assess overfitting and robustness.

3.7 Locality Ablation and Leakage-Control Study

To quantify the contribution of fine-grained geographic information, we conducted a locality ablation study using Random Forest under three settings:

- **Setting A:** structural and regional features only;
- **Setting B:** the same feature set augmented with target-encoded suburb information;
- **Setting C:** the same encoded pipeline as Setting B, but with the suburb labels randomly shuffled before model fitting.

This design answers two questions: *Does suburb-level encoding improve housing price prediction beyond broader structural and regional variables?* and *Does the observed gain remain after a control that disrupts meaningful locality signal?*

4 Results and Discussion

4.1 Extended Baseline Comparison

This section reports the model comparison results and then examines the effect of suburb-level encoding on prediction performance. Table 3 presents the 5-fold cross-validation performance of the evaluated linear, bagging, and boosting models on the log-price scale.

The revised experiments show that boosting models outperform the previously evaluated baselines under the current setting. XGBoost and LightGBM achieve the lowest MAE and RMSE and the highest R^2, while Random Forest remains a strong non-boosting baseline. Linear Regression and Ridge Regression again perform

Table 3 Five-fold cross-validation results on the log-price scale (mean ± standard deviation).

Model	MAE	RMSE	R^2
Linear Regression	0.187 ± 0.004	0.241 ± 0.005	0.736 ± 0.015
Ridge Regression	0.187 ± 0.004	0.241 ± 0.005	0.736 ± 0.015
Lasso Regression	0.189 ± 0.004	0.244 ± 0.005	0.729 ± 0.014
Random Forest	0.152 ± 0.003	0.199 ± 0.003	0.819 ± 0.007
Extra Trees	0.156 ± 0.003	0.205 ± 0.003	0.808 ± 0.008
XGBoost	0.140 ± 0.003	0.185 ± 0.002	0.843 ± 0.003
LightGBM	0.141 ± 0.003	0.185 ± 0.002	0.843 ± 0.005

Table 4 Random Forest suburb ablation and shuffled-suburb control on the log-price scale.

Setting	Train R^2	Test R^2	MAE	RMSE
Without Suburb	0.847	0.815	0.155	0.204
With Suburb	0.871	0.835	0.147	0.193
Shuffled Suburb	0.851	0.810	0.157	0.207

almost identically, whereas Lasso is slightly weaker. This pattern suggests that the Melbourne housing dataset contains non-linear and interaction-driven relationships that are better captured by boosting methods than by linear baselines.

4.2 Impact of Suburb-Level Encoding and Leakage-Control Test

To quantify the effect of suburb-level encoding, we evaluated Random Forest under the locality-ablation and shuffled-suburb control described in Section 3. The results are summarized in Table 4.

Adding suburb information improves hold-out performance, reducing MAE from 0.155 to 0.147 and RMSE from 0.204 to 0.193, while increasing test R^2 from 0.815 to 0.835. The shuffled-suburb control degrades performance to MAE 0.157, RMSE 0.207, and test R^2 0.810. This result shows that the performance gain is not preserved when the locality labels are destroyed. In other words, the model is using meaningful suburb signal rather than benefiting from spurious encoding artifacts.

The train–test gap remains moderate across all three settings. Without suburb, the gap is approximately 0.032; with suburb, it is 0.036; and under the shuffled-suburb control, it is 0.041. These values indicate some overfitting, but not to an extent that invalidates the predictive conclusions. Learning-curve analysis also suggests that the model benefits from additional data, although a residual train–validation gap remains.

4.3 Model Interpretation with SHAP

To move beyond impurity-based feature importance, we used SHAP to interpret the final gradient-boosting model [1]. Figure 3 shows the SHAP summary plot. The most influential variables are `Suburb`, `Distance`, `Landsize`, and `Rooms`, followed by selected region and property-type indicators. The plot also shows a clear directional pattern: higher suburb encoding values and larger land sizes tend to push predictions upward, while larger distance values often push predictions downward. The SHAP plot therefore supports the same substantive conclusion as the ablation results: fine-grained locality information is not only predictive, but central to the model's decisions.

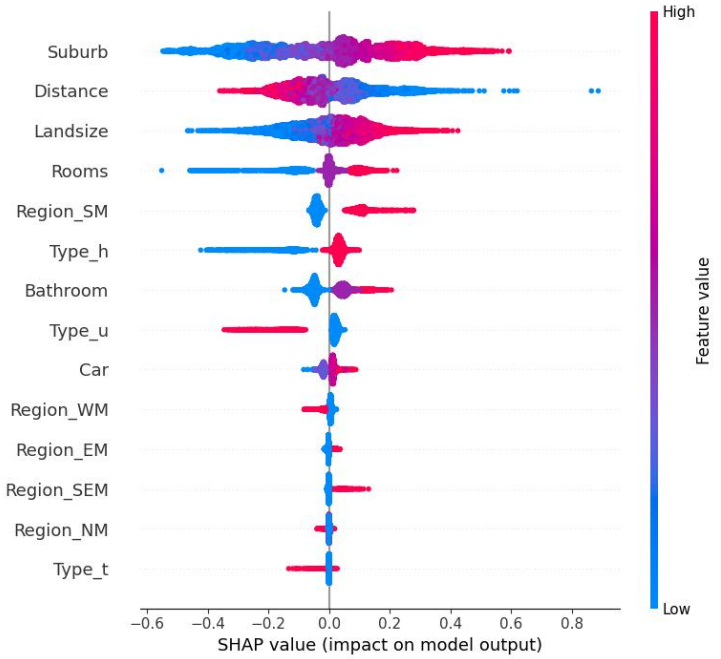


Fig. 3 SHAP summary plot for the final XGBoost model. Suburb, Distance, Landsize, and Rooms are the most influential features, with additional contributions from selected region and property-type indicators.

4.4 Discussion

Taken together, the revised experiments strengthen the main claim of the paper. First, detailed locality encoding improves predictive performance beyond structural and regional features alone. Second, the shuffled-suburb control provides an ex-

plicit check against leakage-like effects in target encoding. Third, stronger boosting baselines outperform Random Forest under the same leakage-safe pipeline, which suggests that the main finding is not tied to a single model family. From an application perspective, this means that housing price prediction systems should treat location encoding as a core design decision and should validate that any gain from encoded locality features survives robustness checks.

5 Conclusion

This paper examined housing price prediction on the Melbourne dataset using a practical machine learning pipeline that combined data cleaning, exploratory analysis, feature engineering, locality encoding, and regression modeling. In the revised final version, we extended the empirical comparison beyond linear and Random Forest baselines by adding Extra Trees, XGBoost, and LightGBM.

The results show two main findings. First, suburb-level encoding improves hold-out prediction performance beyond broader structural and regional variables. Second, stronger boosting models achieve the best overall cross-validation performance under the current setting, with XGBoost and LightGBM outperforming the evaluated linear and bagging baselines. The shuffled-suburb control further shows that the gain from suburb encoding disappears when locality labels are randomized, which supports the claim that the model relies on genuine geographic signal rather than spurious encoding effects.

SHAP analysis also indicates that `Suburb`, `Distance`, `Landsize`, and `Rooms` are the most influential features in the final model. Overall, the revised results show that both model choice and locality encoding matter in housing price prediction, and that robust evaluation should include both strong baselines and explicit controls against leakage.

Future work should evaluate temporal train/test splits to better assess market drift, validate the pipeline on housing datasets from other cities, and compare additional boosting baselines such as CatBoost under the same leakage-safe setting.

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